Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming I part–June 19th, 2023

Polimi ID _____ _____

Surname_____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. Please define the different phases followed in a PRAM computation. (1)

B. Please describe what "Arbitrary CW" means. (1)

C. Please define the efficiency of a parallel algorithm according to the PRAM model (1)

D. Please compare the assumptions behind the Gustafson and the Amdahl laws. (1)

Exercise n. 2

Answer the following questions about algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe how Strassen's algorithm works. (2)

B. Why is the Master Theorem method frequently used to evaluate the asymptotic complexity of a Divide and Conquer algorithm can be computed? (2)

Exercise n. 3

Answer the following questions about the advanced algorithm and briefly explain (without an explanation, the answer will be considered invalid)

A. Please explain why an algorithm could be classified as a Dynamic programming algorithm. (2) (You can skip this question if you completed the first challenge)

B. Describe the treap data structure, how the search is performed, and which complexity has. (2)

Exercise n. 4

Answer the following questions about the amortized analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. Please explain what aggregate amortized analysis is. (2)

B. Please explain why the amortized analysis matters for the analysis of the complexity of the Move to Front algorithm. (2)